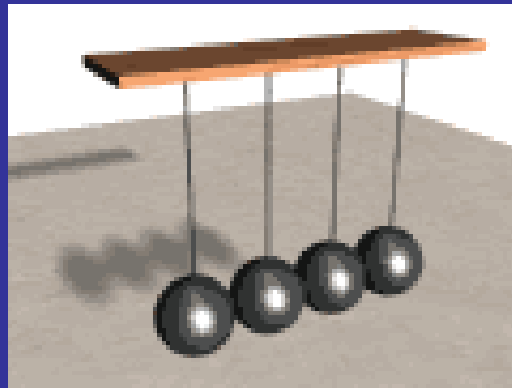
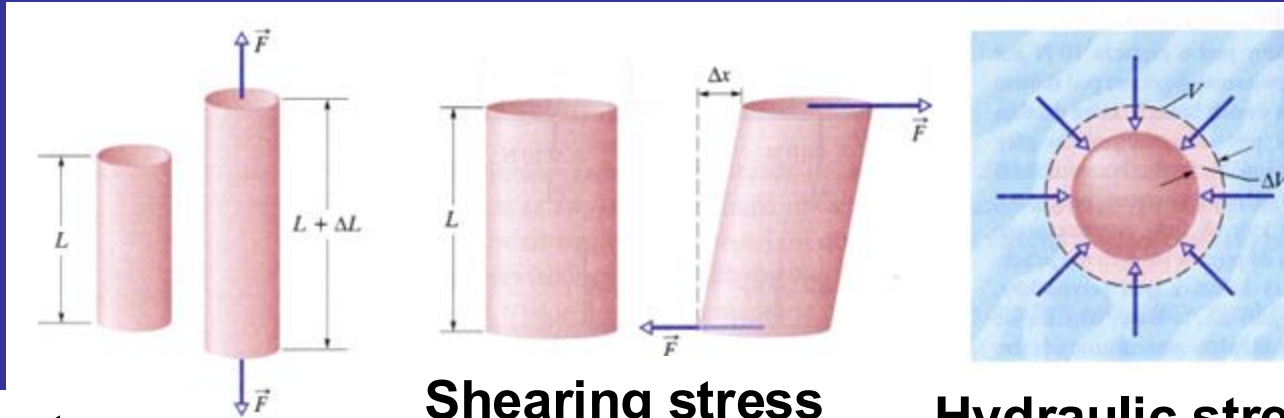


HARMONIC OSCILLATOR



Elastic properties of solid state

**Tensile stress****Shearing stress****Hydraulic stress**

A stress (deforming force per unit area) produces a strain (unit deformation)

$$\text{stress} = (\text{modulus of elasticity}) \times (\text{strain})$$

Provided that the specimen recovers its original dimensions when the stress is removed

Tension and Compression

Stress σ is defined as:

$$\sigma = \frac{F}{S}$$

where F is the magnitude of the force applied perpendicularly to an area S on the object.

The strain, or unit deformation is the dimensionless quantity

$\Delta L/L$ – the fractional change in a length of the specimen

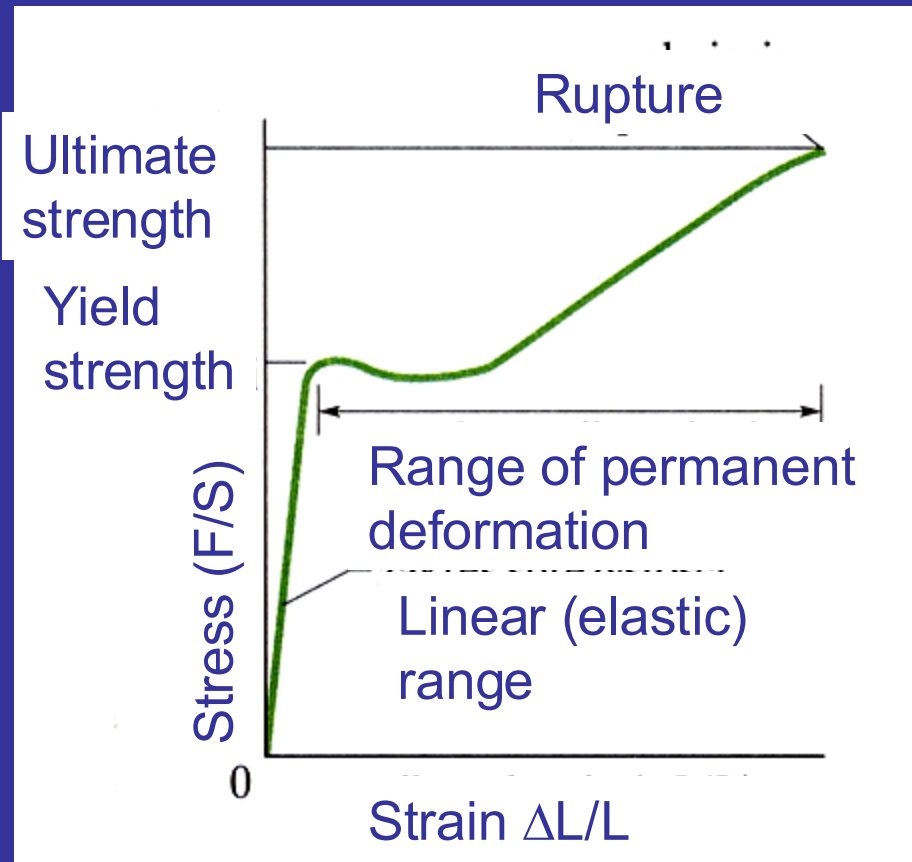
If the specimen is a long rod and the stress does not exceed the yield strength then we have the Hooke's law

$$\frac{F}{S} = E \frac{\Delta L}{L}$$

Tension and Compression

$$\frac{F}{S} = E \frac{\Delta L}{L}$$

E- Young's modulus



Some elastic properties of selected materials of engineering interest

Material	Density ρ (kg/cm ³)	Young's modulus E (10 ⁹ N/m ²)	Ultimate strength (10 ⁶ N/m ²)	Yield strength (10 ⁶ N/m ²)
Steel ^a	7860	200	400	250
Al	2710	70	110	95
Concrete _c	2320	30	40 ^b	-
Bone	1900	9 ^b	170 ^b	-

^a structural steel ASTM-A36, ^b in compression, ^c high strength

Shearing stress

In the case of transversal stress (shearing stress) σ is defined as:

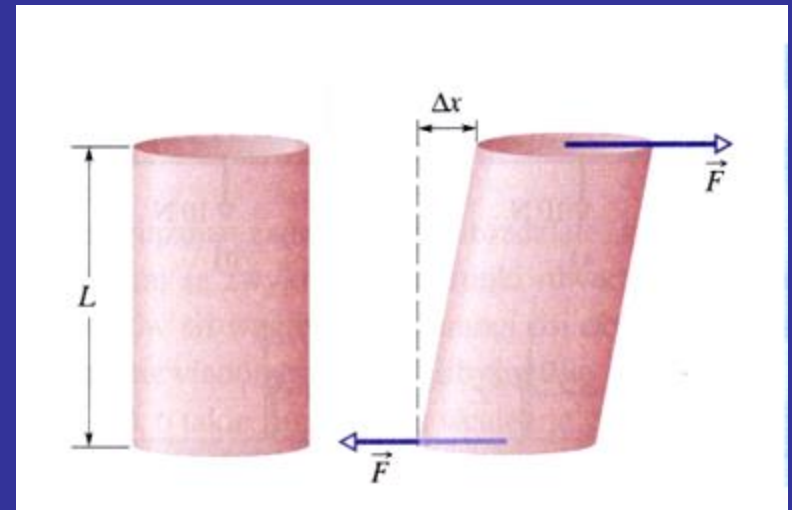
$$\sigma = \frac{F}{S}$$

but the force is parallel to the cross- sectional surface

The strain is the dimensionless ratio $\Delta x/L$

$$\frac{F}{S} = G \frac{\Delta x}{L}$$

shear modulus



Hydraulic stress

The stress is the fluid pressure p on the object

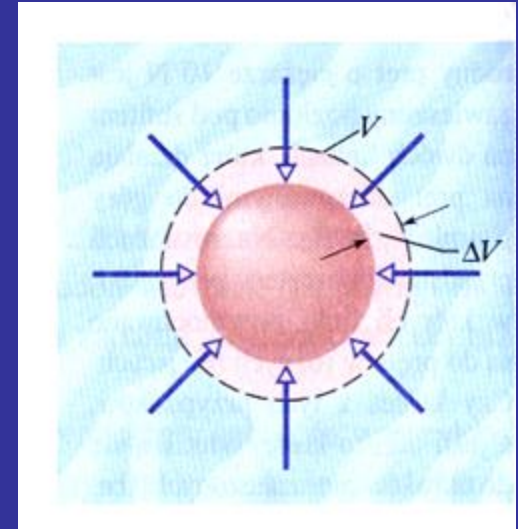
$$p = \frac{F}{S}$$

The unit of pressure is $1 \text{ Pa} = 1 \text{ N/m}^2$

The relative change in volume $\Delta V/V$ is a measure of strain

$$p = B \frac{\Delta V}{V}$$

B – bulk modulus



Example 1 The pressure at the bottom of the Pacific Ocean, at its average depth of about 4000 m, is $4.0 \cdot 10^7 \text{ N/m}^2$. Calculate the fractional compression of a volume $\Delta V/V$ of water related to the pressure and compare it with its value for the solid steel sphere. Bulk modulus of water is $2.2 \cdot 10^9 \text{ N/m}^2$ and for steel - $16 \cdot 10^{10} \text{ N/m}^2$.

Solution:

$$\frac{\Delta V}{V} = \frac{p}{B}$$

water

$$\frac{\Delta V}{V} = \frac{4 \cdot 10^7}{2,2 \cdot 10^9} = 1,8\%$$

steel sphere

$$\frac{\Delta V}{V} = \frac{4 \cdot 10^7}{16 \cdot 10^{10}} = 0,025\%$$

Simple harmonic oscillator

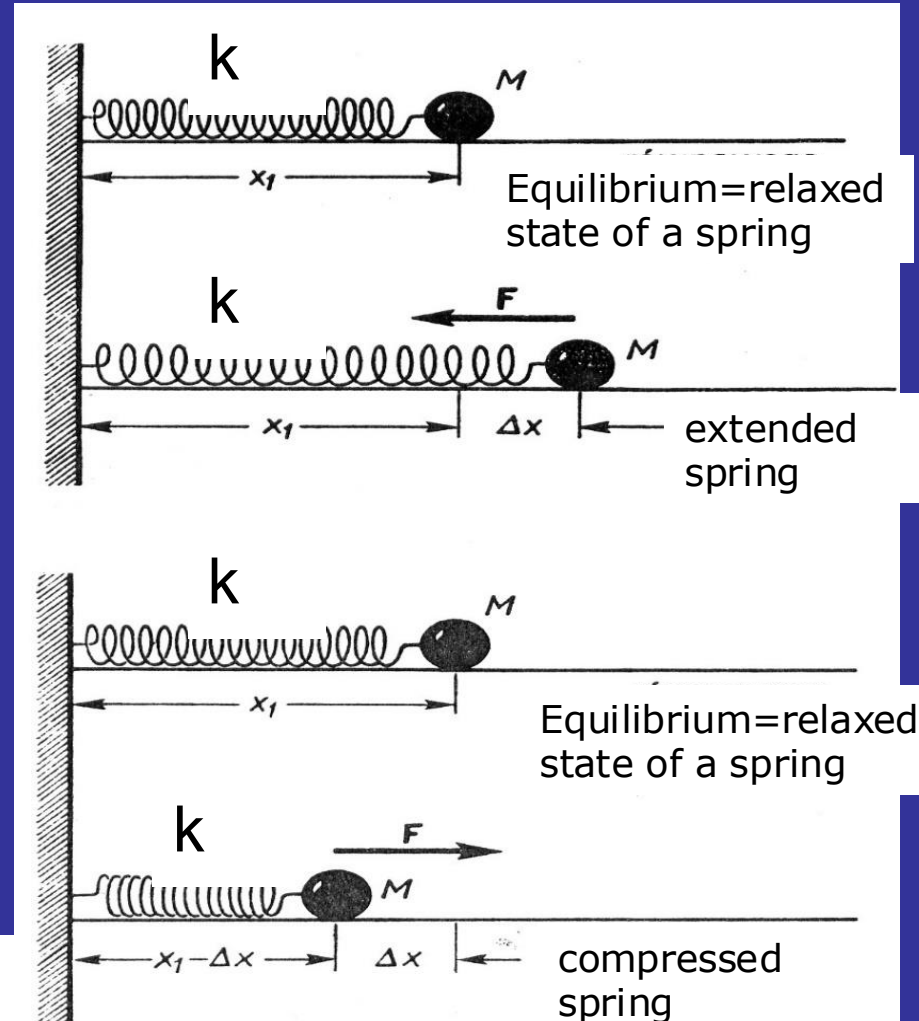
Harmonic force

$$F = -kx$$

- F is proportional to the displacement x from the equilibrium position $x_1=0$
- Pointing towards the equilibrium position

$$\frac{F}{S} = E \frac{\Delta L}{L} \quad \Rightarrow \quad k = ES / L$$

For small displacements x !!!



Equation of motion can be obtained from the second Newton law of motion:

$$F_{\text{wyp}} = ma = m \frac{d^2 x}{dt^2}$$

$$F_{\text{wyp}} = F = -kx$$

comparing and transforming

We obtain the general, differential equation of the harmonic oscillator – second order differential equation with constant coefficients, uniform

$$m \frac{d^2 x}{dt^2} + kx = 0$$

Dividing by m , and assuming that

$$\omega_o = \sqrt{\frac{k}{m}}$$

we get

$$\frac{d^2 x}{dt^2} + \omega_o^2 x = 0$$

Remainder:

Equation of simple harmonic motion SHM

$$m \frac{d^2 x}{dt^2} + kx = 0$$

has been derived from the law of energy conservation

simple harmonic oscillator (without damping and without forcing or driving)

$$\frac{d^2x}{dt^2} + \omega_o^2 x = 0$$


natural angular frequency

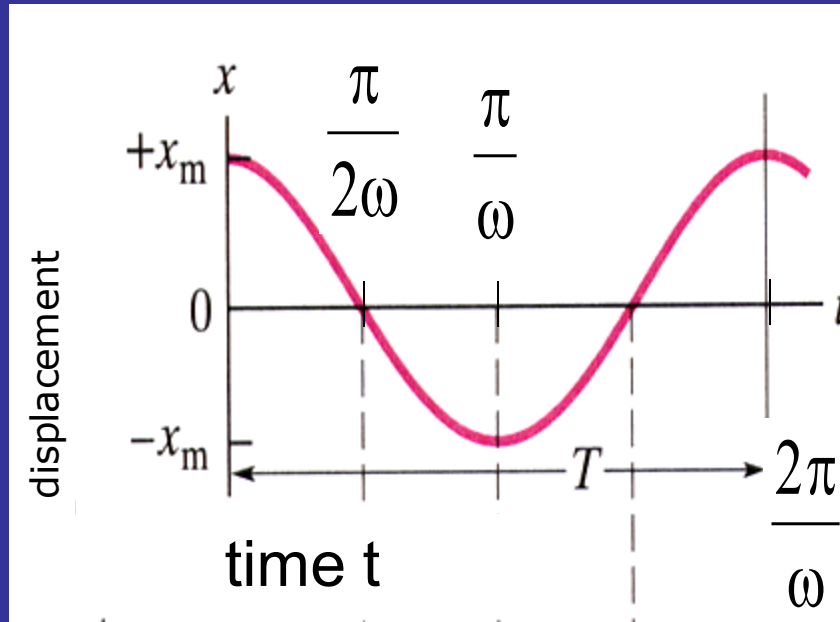
Natural angular frequency depends on the parameters of the oscillator, only. For a system mass m – spring of a given constant k :

$$\omega_o = \sqrt{\frac{k}{m}}$$

From that we can calculate the period T

$$T = \frac{2\pi}{\omega_o} = 2\pi \sqrt{\frac{m}{k}}$$

Solution of simple harmonic oscillator



$$x(t) = x_m \cos(\omega t + \varphi)$$

Displacement
from the
equilibrium

amplitude

Angular
frequency

Initial phase

Amplitude = absolute value of the
maximum displacement

Period T = time interval during
which one full cycle takes
place

$$T = \frac{1}{\nu}$$

frequency = number of
oscillations (cycles) per
second (unit 1Hz)

Velocity

$$v(t) = \frac{dx(t)}{dt} = \frac{d}{dt} (x_m \cos(\omega t + \varphi))$$

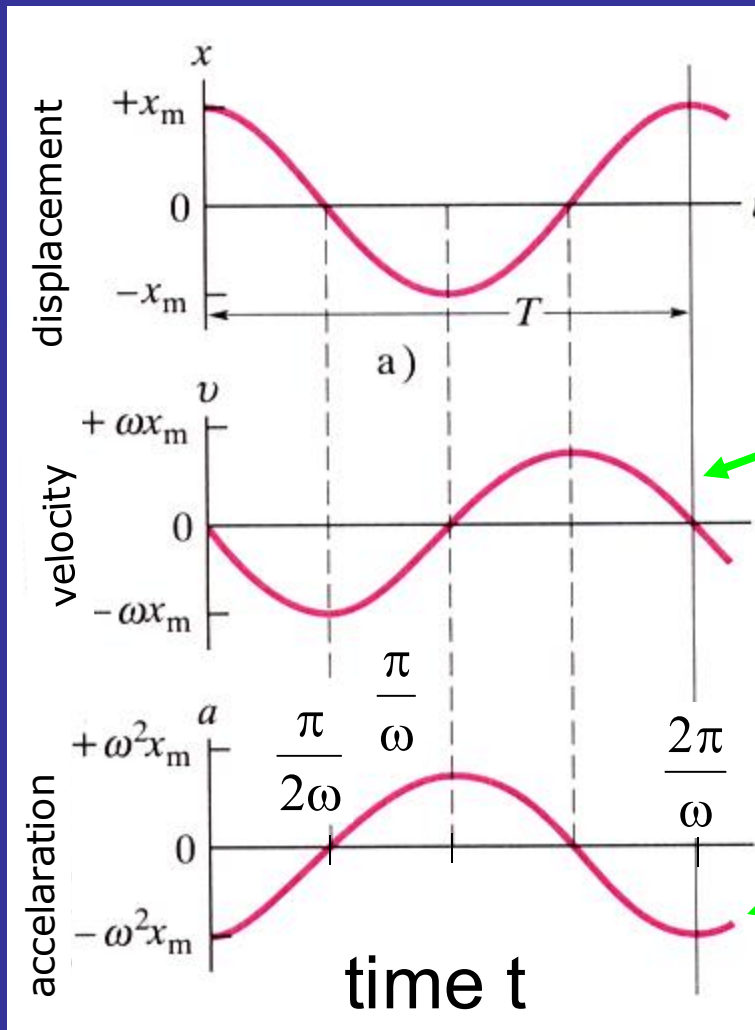
$$v(t) = -x_m \omega \sin(\omega t + \varphi)$$

Acceleration

$$a(t) = \frac{dv(t)}{dt} = \frac{d}{dt} (-x_m \omega \sin(\omega t + \varphi))$$

$$a(t) = -x_m \omega^2 \cos(\omega t + \varphi)$$

$$a = -\omega^2 x$$



Acceleration is proportional to displacement but has an opposite sign

We want to verify whether the proposed function

$$x(t) = x_m \cos(\omega t + \varphi)$$

is a solution of the general differential equation:

$$\frac{d^2x}{dt^2} + \omega_o^2 x = 0$$

We put:

$$\frac{d^2x}{dt^2} = a = -\omega^2 x$$

to the general equation

We get :

$$-\omega^2 x + \omega_o^2 x = 0$$

from which

Angular frequency ω of simple harmonic motion SHM is equal to the natural frequency ω_o

$$\omega = \omega_o$$

Force in simple harmonic motion SHM

acceleration

$$a = -\omega^2 x$$

From the second law of dynamics:

$$F_{\text{wyp}} = ma = -m\omega^2 x$$

but harmonic force:

$$F = -kx$$


$$k = m\omega^2$$

We know that

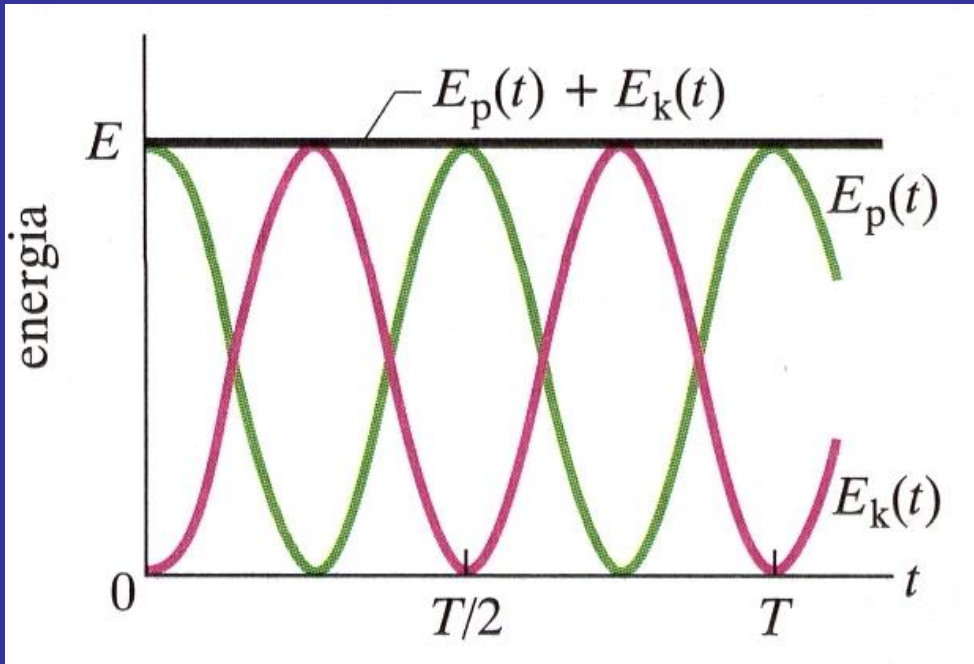
$$\omega_o = \sqrt{\frac{k}{m}}$$

then again:

$$\omega = \omega_o$$

Energy in SHM

Elastic potential energy



$$E_p = k \frac{x^2}{2}$$

$$E_p = \frac{1}{2} k x_m^2 \cos^2(\omega t + \varphi)$$

Kinetic energy

$$E_k = \frac{1}{2} m \omega^2 x_m^2 \sin^2(\omega t + \varphi)$$

$$E_k = m \frac{v^2}{2}$$

Total energy in SHM

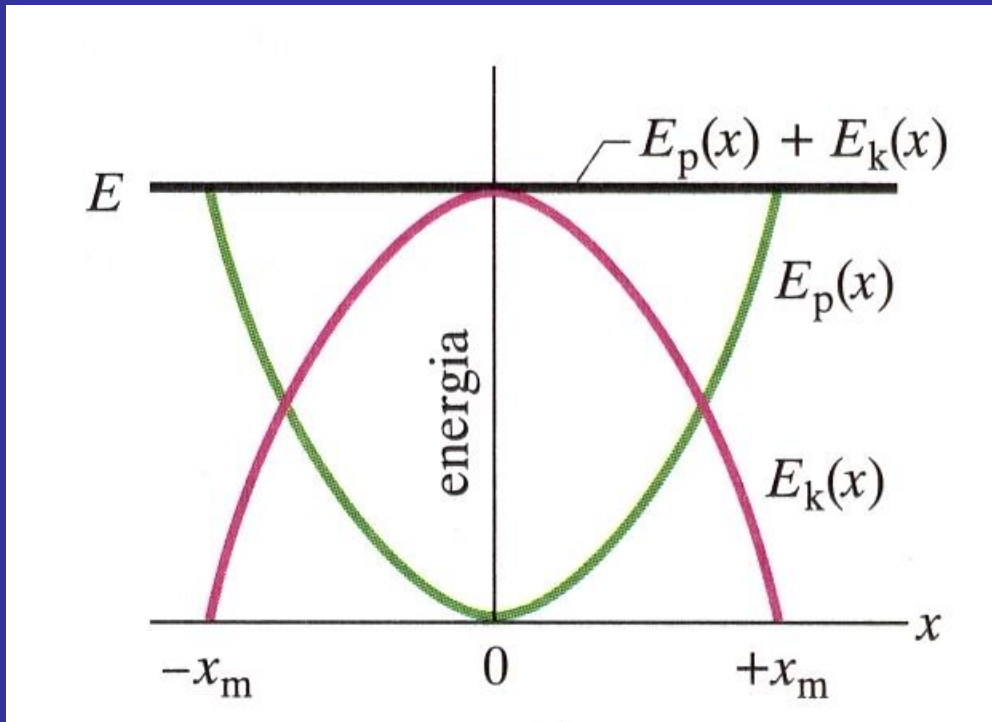
$$E = E_p + E_k = \frac{1}{2}kx_m^2 \cos^2(\omega t + \varphi) + \frac{1}{2}m\omega^2 x_m^2 \sin^2(\omega t + \varphi)$$

but $k = m\omega^2$ hence

$$E = \frac{1}{2}kx_m^2 (\cos^2(\omega t + \varphi) + \sin^2(\omega t + \varphi)) = \frac{1}{2}kx_m^2 = \text{const}$$

Total mechanical energy in SHM is conserved

Dependence of the energy of a simple harmonic oscillator on the displacement from the equilibrium position



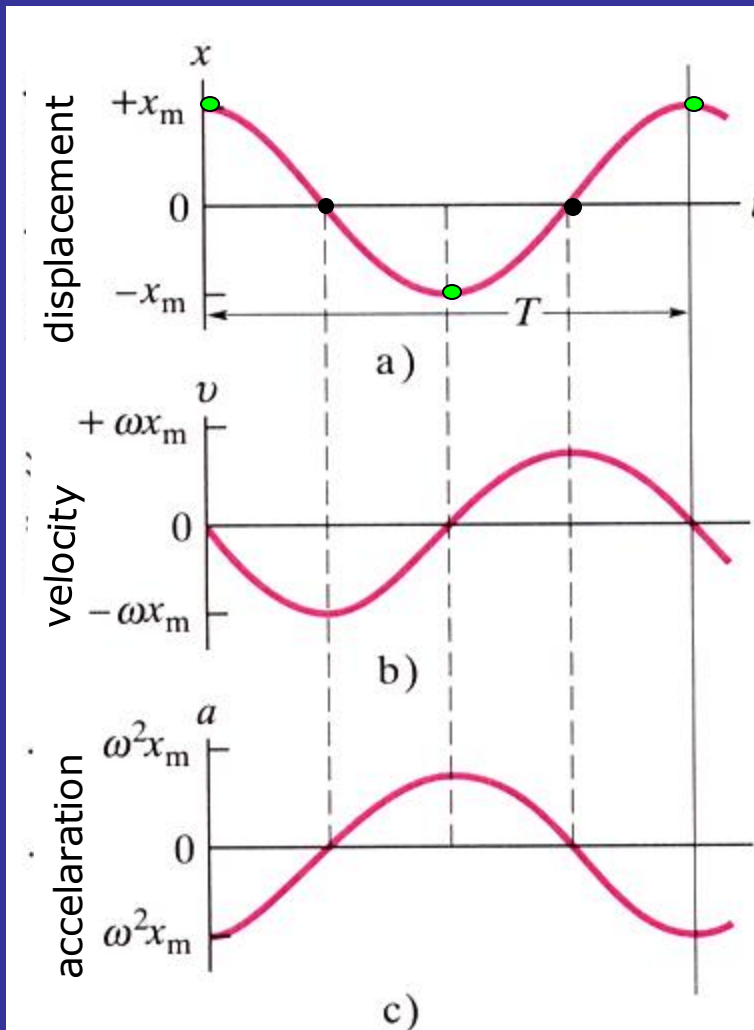
Inversed parabola

$$E_p = k \frac{x^2}{2} \quad \text{Parabolic dependence}$$

$$E_k = E - E_p$$

$$E_k = \frac{1}{2} k x_m^2 - \frac{1}{2} k x^2$$

$$E_k = \frac{1}{2} k (x_m^2 - x^2)$$



Passing through the equilibrium:

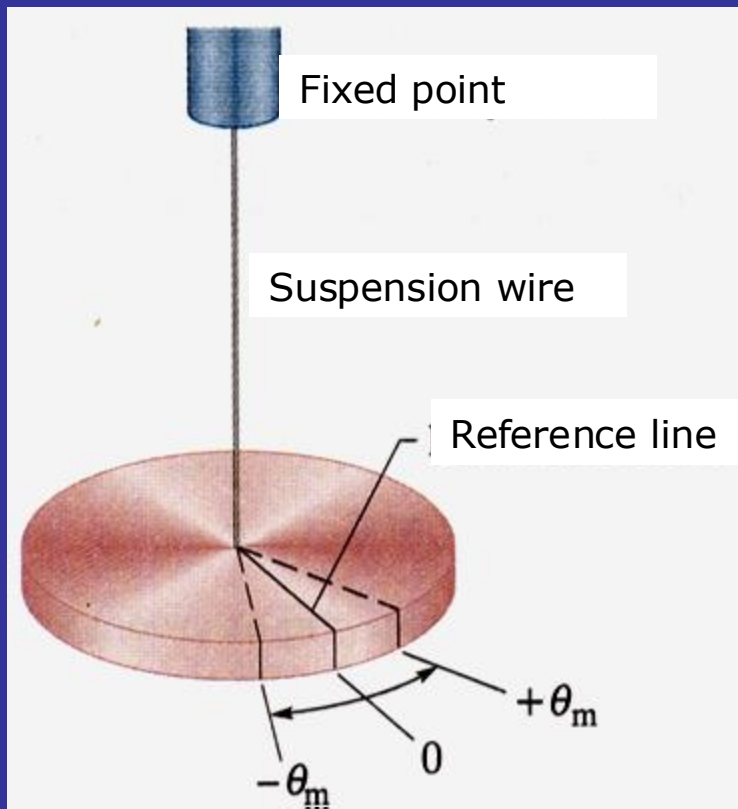
- velocity is max
- acceleration is 0
- force is 0
- kinetic energy is max

At the position corresponding to the amplitude:

- velocity is 0 and changes its sign
- acceleration is max
- force is max
- potential energy is max

EXAMPLES OF HARMONIC OSCILLATORS

Torsion pendulum



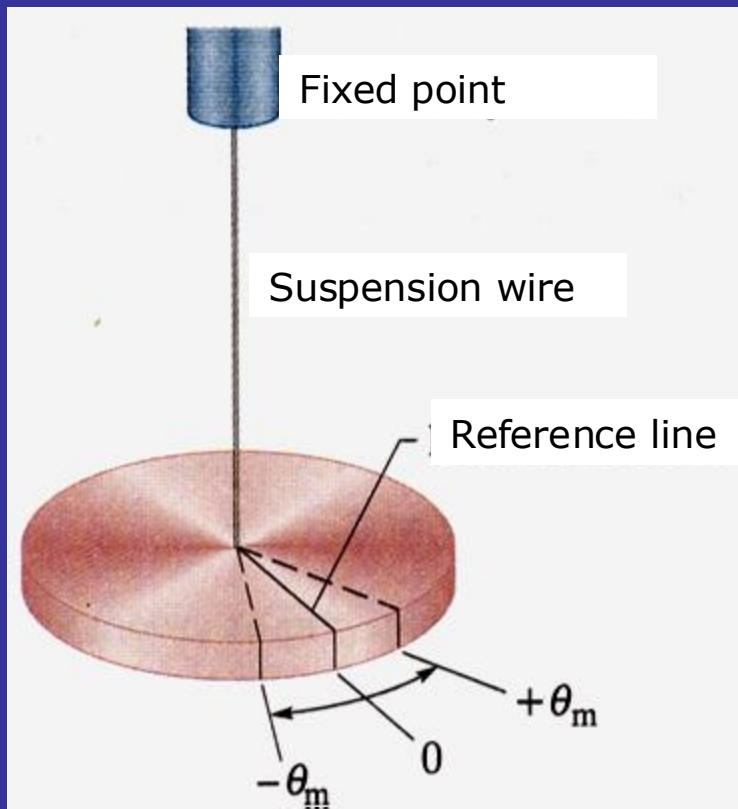
$$\tau = -\kappa\theta$$

Torsion constant κ depends on the length, diameter, and material of the wire

$$I \frac{d^2\theta}{dt^2} = \tau$$

Equation of motion from the second law of dynamics for rotational motion

Torsion pendulum



$$I \frac{d^2\theta}{dt^2} = -\kappa\theta \rightarrow I \frac{d^2\theta}{dt^2} + \kappa\theta = 0$$

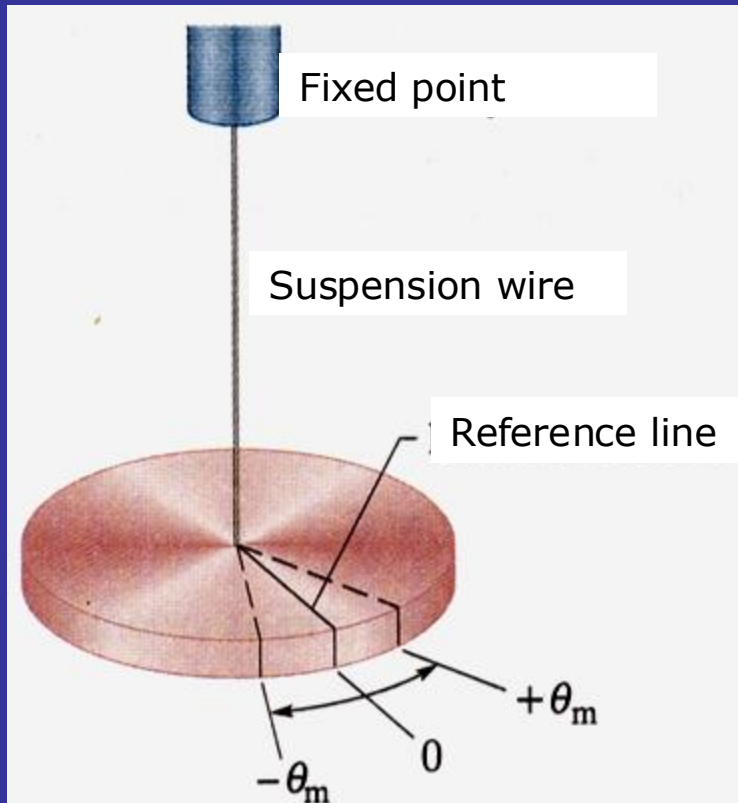
$$\frac{d^2\theta}{dt^2} + \frac{\kappa}{I}\theta = 0$$

$$\frac{d^2\theta}{dt^2} + \omega_o^2\theta = 0$$

Equation of harmonic oscillator

$$\omega_o^2 = \frac{\kappa}{I}$$

Torsion pendulum



$$\omega_o^2 = \frac{\kappa}{I}$$

$$T = \frac{2\pi}{\omega_o} = 2\pi \sqrt{\frac{I}{\kappa}}$$

A red arrow points from the ω_o term in the denominator of the second equation to the text 'Period of torsion pendulum' below.

Period of torsion pendulum

Torsion pendulum can be practically applied in the determination of moment of inertia of rigid bodies of irregular shape

Example 2. Figure shows a thin rod whose length $L=12.4$ cm and whose mass $m=135$ g suspended at its midpoint from a long wire. Its period T_a of torsional oscillations is measured to be 2.53 s. An irregular shaped object, which we call object X, is then hung from the same wire and its period T_b is 4.76 s. What is the rotational inertia of object X about its suspension axis?

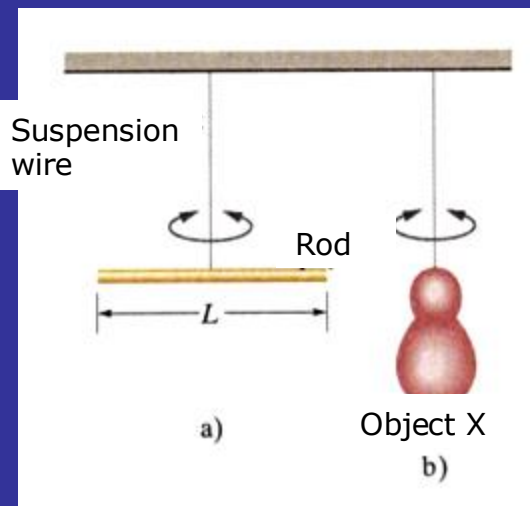
Given:

$$L=12.4 \text{ cm}=0.124 \text{ m}$$

$$m=135 \text{ g}=0.135 \text{ kg}$$

$$T_a=2.53 \text{ s}$$

$$T_b=4.76 \text{ s}$$



Calculated:

$$I_b$$

Solution:

Period of torsion pendulum with a rod:

$$T_a = 2\pi\sqrt{\frac{I_a}{\kappa}}$$

Period of torsion pendulum with an unknown rigid body X:

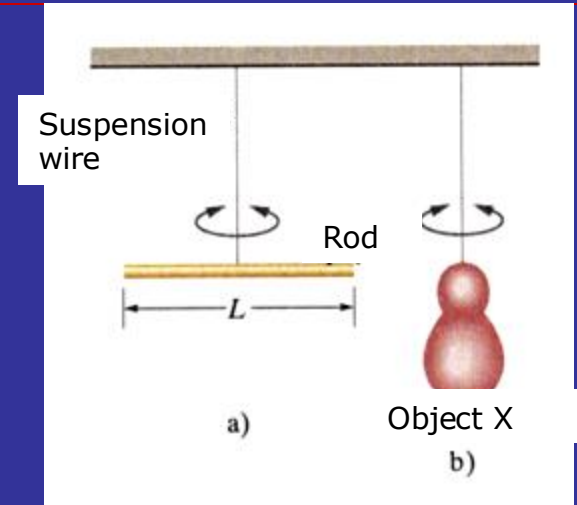
$$T_b = 2\pi\sqrt{\frac{I_b}{\kappa}}$$

Moment of inertia of X:

$$I_b = I_a \frac{T_b^2}{T_a^2} = \frac{1}{12} mL^2 \frac{T_b^2}{T_a^2}$$

Response:

$$I_b = 6,12 \cdot 10^{-4} \text{ kg} \cdot \text{m}^2$$



Simple pendulum

Torque of gravitational force is responsible for this oscillatory behaviour:

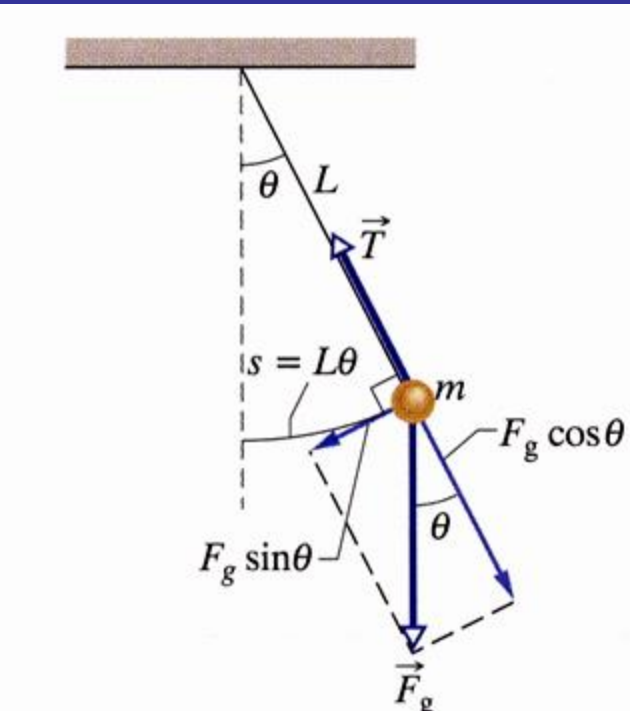
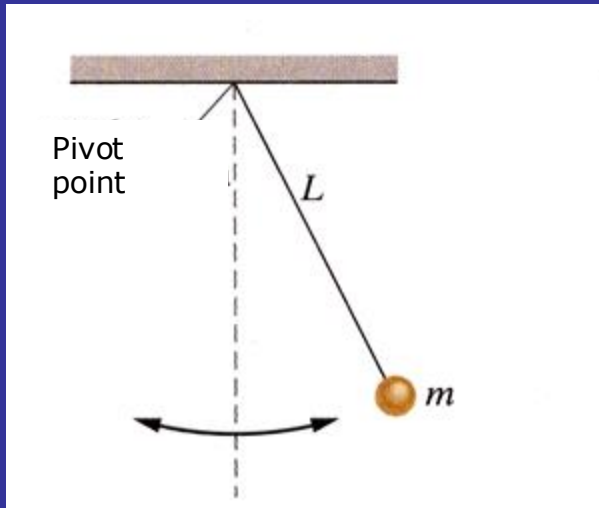
$$\tau = -L(F_g \sin \theta) = -Lmg \sin \theta$$

„minus” sign indicates that the torque causes a decrease of angle θ

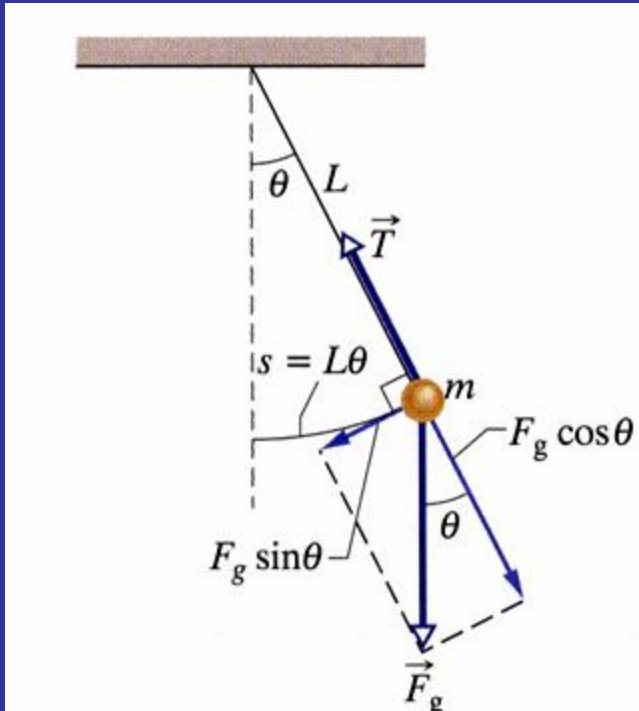
From the second law of rotational motion:

I-moment of inertia with respect to the pivot point

$$\tau = I\epsilon = I \frac{d^2\theta}{dt^2}$$



Simple pendulum



We assume that the angle θ is small
thus $\sin \theta \approx \theta$:

$$\tau = -Lmg\theta$$

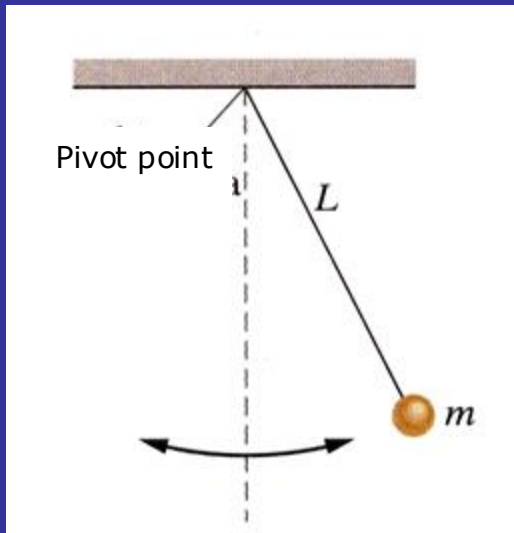
Equation of harmonic oscillator:

$$I \frac{d^2\theta}{dt^2} + Lmg \theta = 0$$

Natural angular frequency:

$$\omega_o = \sqrt{\frac{Lmg}{I}}$$

Simple pendulum



$$\omega_o = \sqrt{\frac{Lmg}{I}}$$

but

$$I = mL^2$$



$$\omega_o = \sqrt{\frac{g}{L}}$$



Natural angular frequency
does not depend on mass

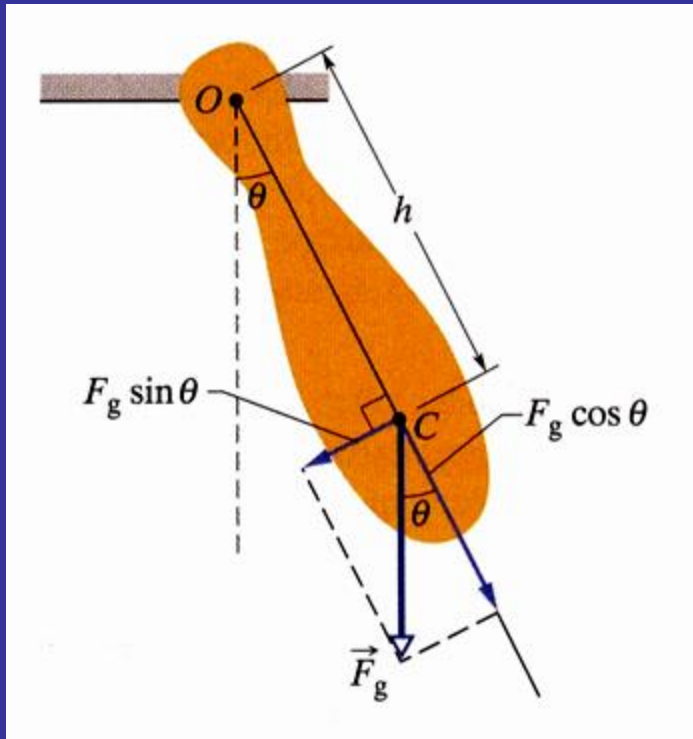
Period T does not depend on mass

$$T = 2\pi \sqrt{\frac{L}{g}}$$

It is true for small amplitude of oscillations

Physical pendulum

Any rigid body in the oscillatory motion constitutes a physical pendulum



$$\tau = -mgh \sin \theta$$

$$I_O \frac{d^2\theta}{dt^2} = -mgh \sin \theta$$

for small angles θ

$$I_O \frac{d^2\theta}{dt^2} + mgh \theta = 0$$

$$\omega_o = \sqrt{\frac{mgh}{I_O}} = \sqrt{\frac{mgh}{I_{cm} + mh^2}}$$

Physical pendulum

Physical pendulum is used to determine experimentally the free-fall acceleration g at different locations on Earth's surface and not only

$$T = 2\pi \sqrt{\frac{I_{\text{cm}} + mh^2}{mgh}}$$

Example 3. A meter stick swings about a pivot point O at one end, at a distance h from the stick's center of mass C , as shown below. Measuring the period of the physical pendulum T , find g at this point.

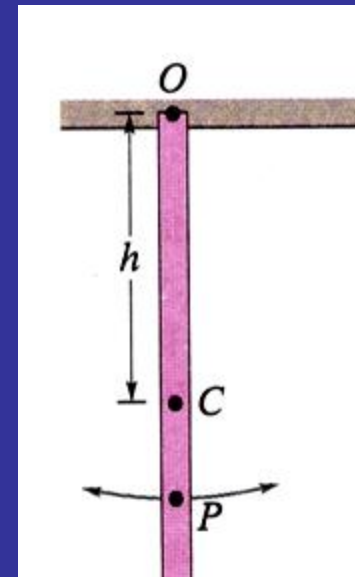
Solution:

$$I_{\text{cm}} = \frac{1}{12} mL^2$$

$$h = \frac{L}{2}$$

$$T = 2\pi \sqrt{\frac{\frac{1}{3} mL^2}{mg \frac{L}{2}}}$$

$$g = \frac{8\pi^2 L}{3T^2}$$



Physical pendulum

Corresponding to any physical pendulum that oscillates about a given pivot point O with period T , there exists a simple pendulum of length L_o with the same period T . Length L_o is called a reduced length of the physical pendulum. Point located at a distance L_o from the pivot point O is called the center of oscillation of the physical pendulum for the given suspension point.

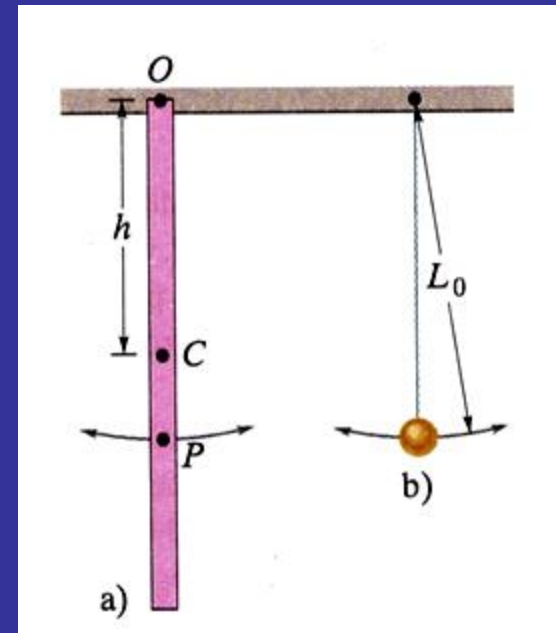
Example 4. Find the reduced length of the pendulum of the previous example and determine the center of oscillation.

$$T = 2\pi\sqrt{\frac{\frac{2}{3}L}{g}}$$

hence

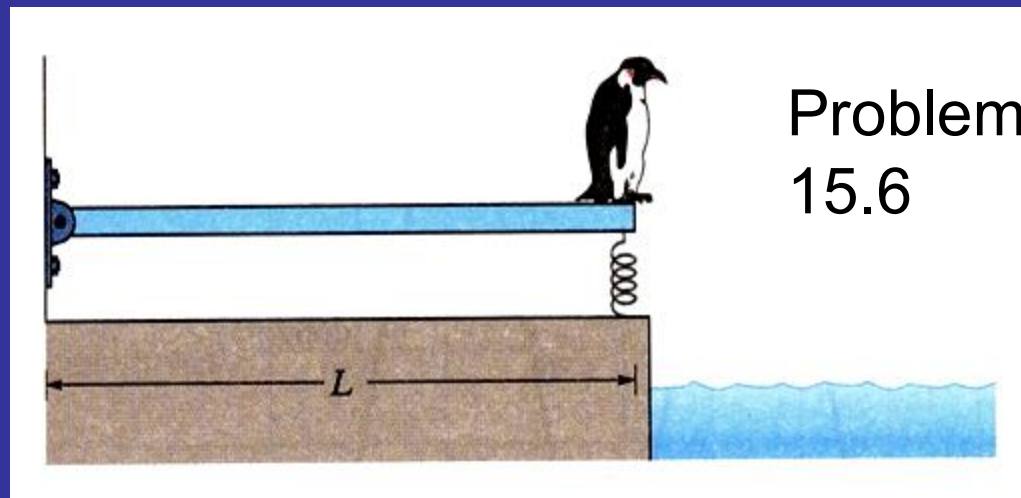
$$L_o = \frac{2}{3}L$$

Center of oscillation is at the point P

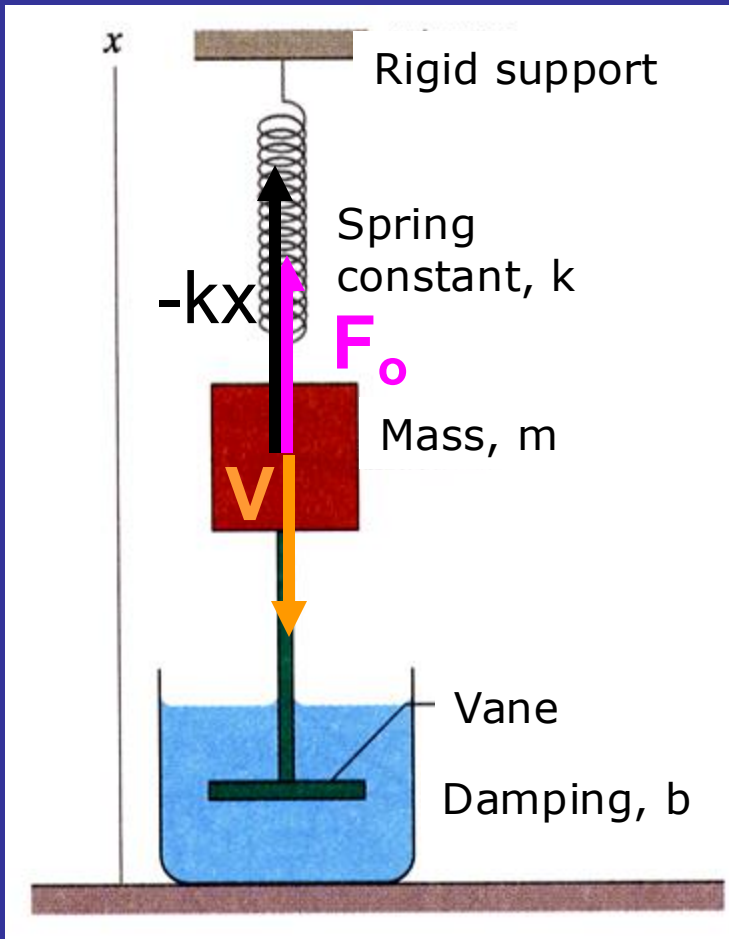


HOMework 8.1

Drawing shows a simple but realistic model of a competition diving board. The board section beyond the fulcrum is treated as a stiff rod of length L that can rotate about a hinge at the fulcrum, compressing an (imaginary) spring under the board free end. The board length is $L=2\text{m}$ and its mass is $m=12\text{ kg}$; spring constant k amounts to 1300 N/m . When the diver (penguin) lands at the free end of the board, it start oscillating with a small amplitude. We assume that the board is stiff enough not to bend. Find the period of oscillations T .



Damped harmonic oscillator



Drag force= Stokes' force

$$F_o = -bv$$

damping constant

net force

$$F_w = -bv - kx$$

from the second law
of motion

$$ma = -bv - kx$$

hence

$$m \frac{d^2 x}{dt^2} = -bv - kx$$

General equation of harmonic oscillator with damping

$$m \frac{d^2 x}{dt^2} + b \frac{dx}{dt} + kx = 0$$

$$\frac{d^2 x}{dt^2} + \frac{b}{m} \frac{dx}{dt} + \frac{k}{m} x = 0$$



$$\frac{d^2 x}{dt^2} + \frac{1}{\tau} \frac{dx}{dt} + \omega_0^2 x = 0$$

relaxation time

$$\beta = \frac{b}{2m}$$

$$\tau = \frac{1}{2\beta} = \frac{m}{b}$$

or

$$\frac{d^2 x}{dt^2} + 2\beta \frac{dx}{dt} + \omega_0^2 x = 0$$

Coefficient of damping

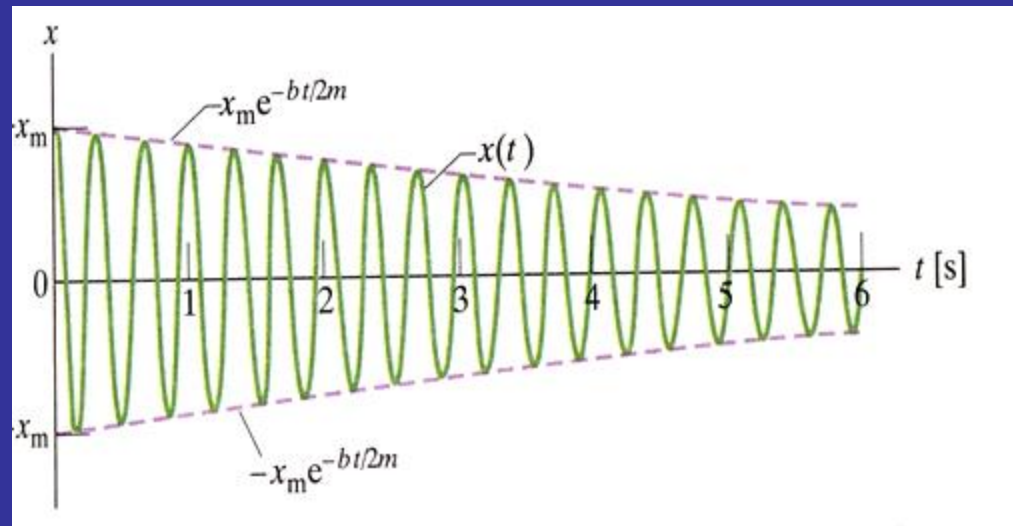
Solution of the damped harmonic oscillator

$$\frac{d^2x}{dt^2} + 2\beta \frac{dx}{dt} + \omega_o^2 x = 0$$

For small damping coefficient, we propose periodic solution with the amplitude of oscillations decreasing exponentially with time,

$$x(t) = e^{-\beta t} z(t)$$

while $z(t)$ is a solution of simple harmonic oscillator



We verify that the following function $x(t) = e^{-\beta t} z(t)$ is a solution of differential equation

$$\frac{d^2 x}{dt^2} + 2\beta \frac{dx}{dt} + \omega_o^2 x = 0$$

We assume:

$$\frac{dx}{dt} = -\beta e^{-\beta t} z(t) + e^{-\beta t} \dot{z}(t)$$

$$\dot{z}(t) = \frac{dz}{dt}$$

$$\ddot{z}(t) = \frac{d^2 z}{dt^2}$$

$$\frac{d^2 x}{dt^2} = \beta^2 e^{-\beta t} z - 2\beta e^{-\beta t} \dot{z}(t) + e^{-\beta t} \ddot{z}$$

Very useful theorem:

$$(fg)'' = f''g + 2f'g' + fg''$$

$$\ddot{z} + \underbrace{(\omega_o^2 - \beta^2)}_{\omega^2} z = 0$$

It is the equation of simple harmonic oscillator if

$$\omega^2 = \omega_o^2 - \beta^2$$

then

$$z(t) = A \cos(\omega t + \phi)$$

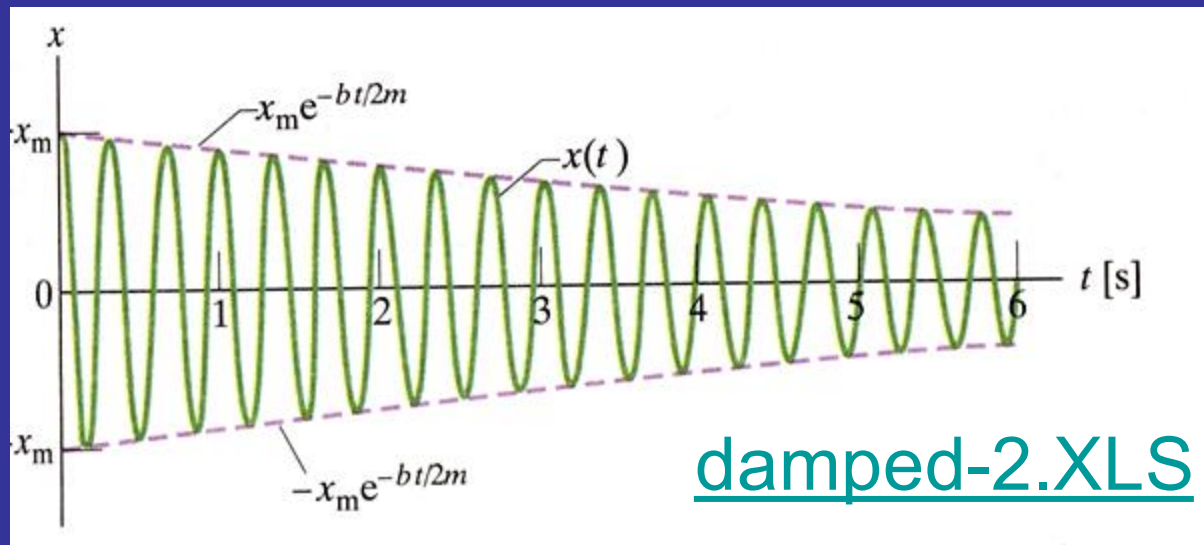
Solution of harmonic oscillator with damping

$$x(t) = x_m \exp\left(\frac{-bt}{2m}\right) \cos(\omega t + \phi)$$

Amplitude depends on time

$$\omega^2 = \omega_o^2 - \beta^2 = \omega_o^2 - \left(\frac{b}{2m}\right)^2$$

Angular frequency is different from the natural frequency and depends on damping



[damped-2.XLS](#)

Solution of the harmonic oscillator with damping in periodic form

$$x(t) = A \exp(-\beta t) \cos(\omega t + \phi)$$

is possible in the case of small damping, only $\beta < \omega_0$

because

$$\omega^2 = \omega_0^2 - \beta^2 > 0$$

When $\beta = \omega_0$ we deal with the critical case

For $\beta > \omega_0$ we obtain the aperiodic solutions

Logarithmic decrement of damping

$$A(t) = x_m \exp\left(\frac{-bt}{2m}\right) = A_o \exp\left(\frac{-bt}{2m}\right)$$

$$A_o = A(t=0)$$

Logarithmic decrement of damping Λ is the natural logarithm of the ratio of the consecutive amplitudes

$$\Lambda = \ln \frac{A_n}{A_{n+1}}$$

$$\Lambda = -\ln \frac{A(t+T)}{A(t)} = -\ln \frac{e^{-\beta(t+T)}}{e^{-\beta t}} = \beta T$$

$$\beta = \frac{b}{2m}$$

damping
coefficient



HOMework 8.2

Think about the interpretation of relaxation time. Calculate how many times the amplitude of damped harmonic oscillator will decrease with respect to the initial amplitude within the time interval corresponding to the relaxation time.

Loss of energy and the Q factor

Q-factor of the oscillator is defined as:

$$Q = 2\pi \frac{\text{energy stored}}{\text{energy dissipated in } T}$$

For a weakly damped harmonic oscillator

$$Q \approx \omega_o \tau$$

Quantity $\omega_o \tau$ or Q is a measure of the lack of damping of the oscillator. Big $\omega_o \tau$ or big Q indicates, that the oscillator is weakly damped, e.g., for a piano string $Q \approx 10^3$, the excited atom $Q \approx 10^7$

HOMework 8.3

How long does it take for the energy of damped harmonic oscillator to decrease to e^{-1} of its initial value? How many full cycles the oscillator will complete within this time interval?

Forced harmonic oscillator - resonance

We assume periodic force driving a damped oscillator in a form:

$$F(t) = F_0 \sin(\omega t)$$



Angular frequency of a driving force

Differential equation of the harmonic oscillator with damping and periodic driving force:

$$m \frac{d^2 x}{dt^2} + b \frac{dx}{dt} + kx = F_0 \sin(\omega t)$$

Forced harmonic oscillator, cont.

dividing by a mass:

$$\frac{d^2x}{dt^2} + \frac{b}{m} \frac{dx}{dt} + \frac{k}{m} x = \frac{F_0}{m} \sin(\omega t)$$

and introducing new constants:

$$\frac{d^2x}{dt^2} + \frac{1}{\tau} \frac{dx}{dt} + \omega_0^2 x = \alpha_0 \sin(\omega t)$$

In a steady state, the oscillations take place with the angular frequency of the driving force, ω

Forced harmonic oscillator, cont.

Solution of the equation:

$$\frac{d^2x}{dt^2} + \frac{1}{\tau} \frac{dx}{dt} + \omega_0^2 x = \alpha_0 \sin(\omega t)$$

is given by:

$$x(t) = x_0(\omega) \sin(\omega t + \varphi(\omega))$$

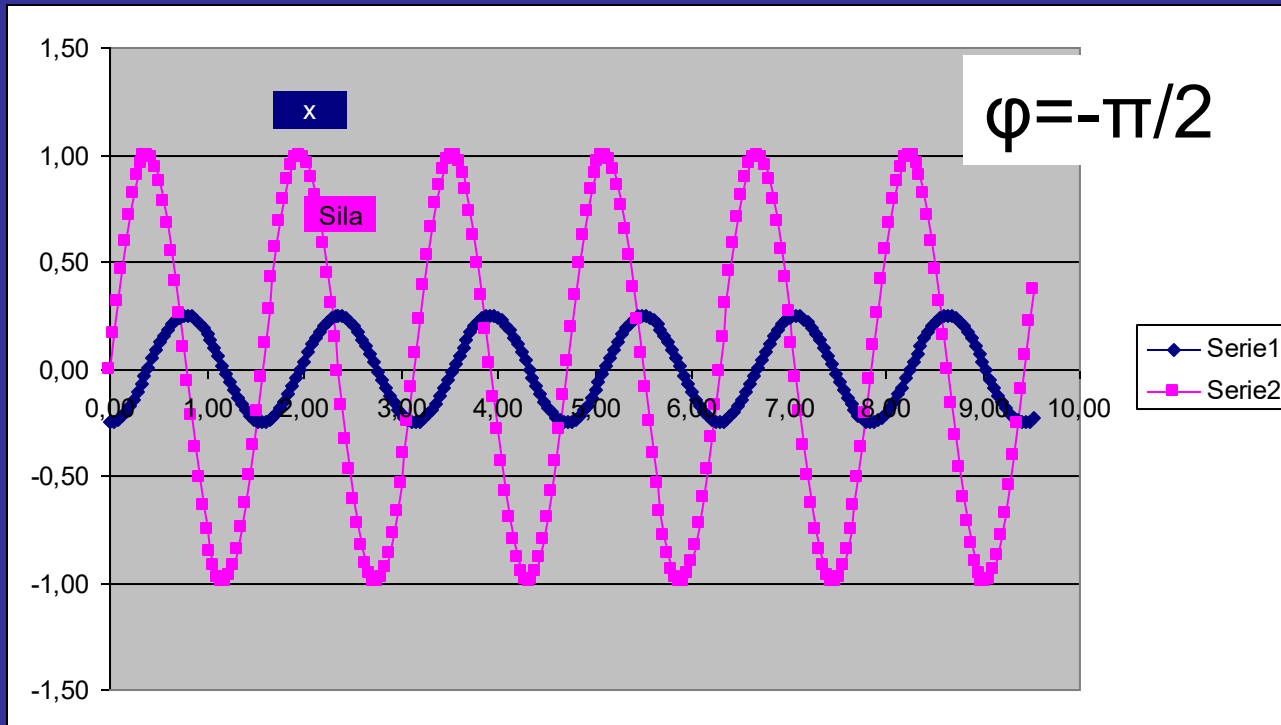
We obtain non-decaying oscillations of the amplitude independent of time, as for a simple harmonic oscillator, but:

- ✓ the amplitude x_0 is a function of driving angular frequency
- ✓ phase shift is not an arbitrary constant but is strictly determined by the driving angular frequency

Phase shift $\varphi(\omega)$ informs us about an angle by which the maximum in displacement advances the maximum in the driving force F .

$$x(t) = x_o(\omega) \sin(\omega t + \varphi(\omega))$$

$$F(t) = F_o \sin(\omega t)$$



We can show that:

the amplitude of oscillations

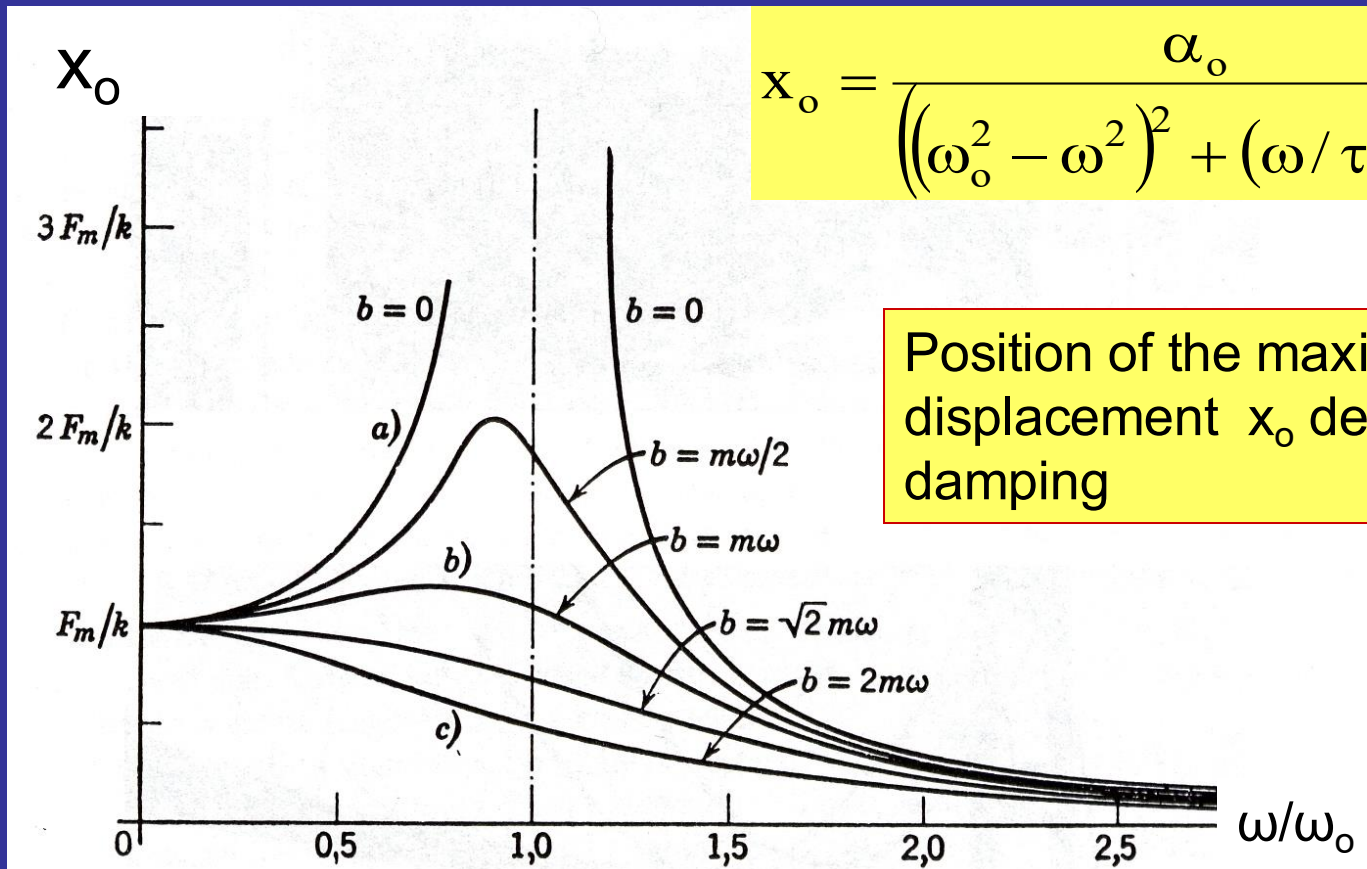
$$x_o = \frac{\alpha_o}{\left(\left(\omega_o^2 - \omega^2 \right)^2 + (\omega / \tau)^2 \right)^{1/2}}$$

phase shift

$$\operatorname{tg} \varphi = - \frac{\omega / \tau}{\omega_o^2 - \omega^2}$$

depend on the angular frequency of the driving force

Resonance appears when the amplitude reaches a maximum value what in practice occurs when the frequency of the driving force approaches that of the natural frequency



$$X_o = \frac{\alpha_o}{\left((\omega_o^2 - \omega^2)^2 + (\omega/\tau)^2 \right)^{1/2}}$$

Position of the maximum of displacement x_o depends on damping

Example 5. Find the condition of the resonance in the case of:

a) maximum amplitude of displacement $x_{\max}$,

b) maximum amplitude of velocity $v_{\max}$.

Solution:

a)

$$x_o = \frac{\alpha_o}{\left((\omega_o^2 - \omega^2)^2 + (\omega/\tau)^2 \right)^{1/2}}$$

$$\frac{dx_o}{d\omega} = ?$$

You have to find the minimum of the denominator

$$f(\omega) = (\omega_o^2 - \omega^2)^2 + (\omega/\tau)^2$$

$$\frac{d}{d\omega} f(\omega) = 2(\omega_o^2 - \omega^2)(-2\omega) + 2(\omega/\tau^2)$$

$$\frac{d}{d\omega} f(\omega_{\text{rez}}) = 0$$

for

$$\omega_{\text{rez}} = \sqrt{\omega_o^2 - \frac{1}{2\tau^2}}$$

$$\omega_{\text{rez}} = \sqrt{\omega_o^2 - \frac{1}{2\tau^2}}$$

Resonant frequency in the case (a) depends on relaxation time

When $\tau \rightarrow \infty$ i.e. no damping, then $\omega_{\text{rez}} = \omega_o$

Amplitude of
oscillations

$$x_o(\omega_{\text{rez}}) = \frac{\alpha_o}{\left(\left(\omega_o^2 - \omega^2 \right)^2 + \underbrace{\left(\omega / \tau \right)^2}_{\rightarrow 0} \right)^{1/2}} \rightarrow \infty$$

$\rightarrow 0$

b) amplitude of velocity

$$x(t) = x_o(\omega) \sin(\omega t + \varphi(\omega))$$

$$v(t) = \frac{d}{dt} x(t) = \omega x_o(\omega) \cos(\omega t + \varphi(\omega))$$

$$V_{\max} = \omega X_o(\omega)$$

$$V_{\max} = \frac{\omega \alpha_o}{\left(\left(\omega_o^2 - \omega^2 \right)^2 + (\omega / \tau)^2 \right)^{1/2}}$$

$$\frac{dV_{\max}}{d\omega} = ?$$

HOMEWORK 8.4

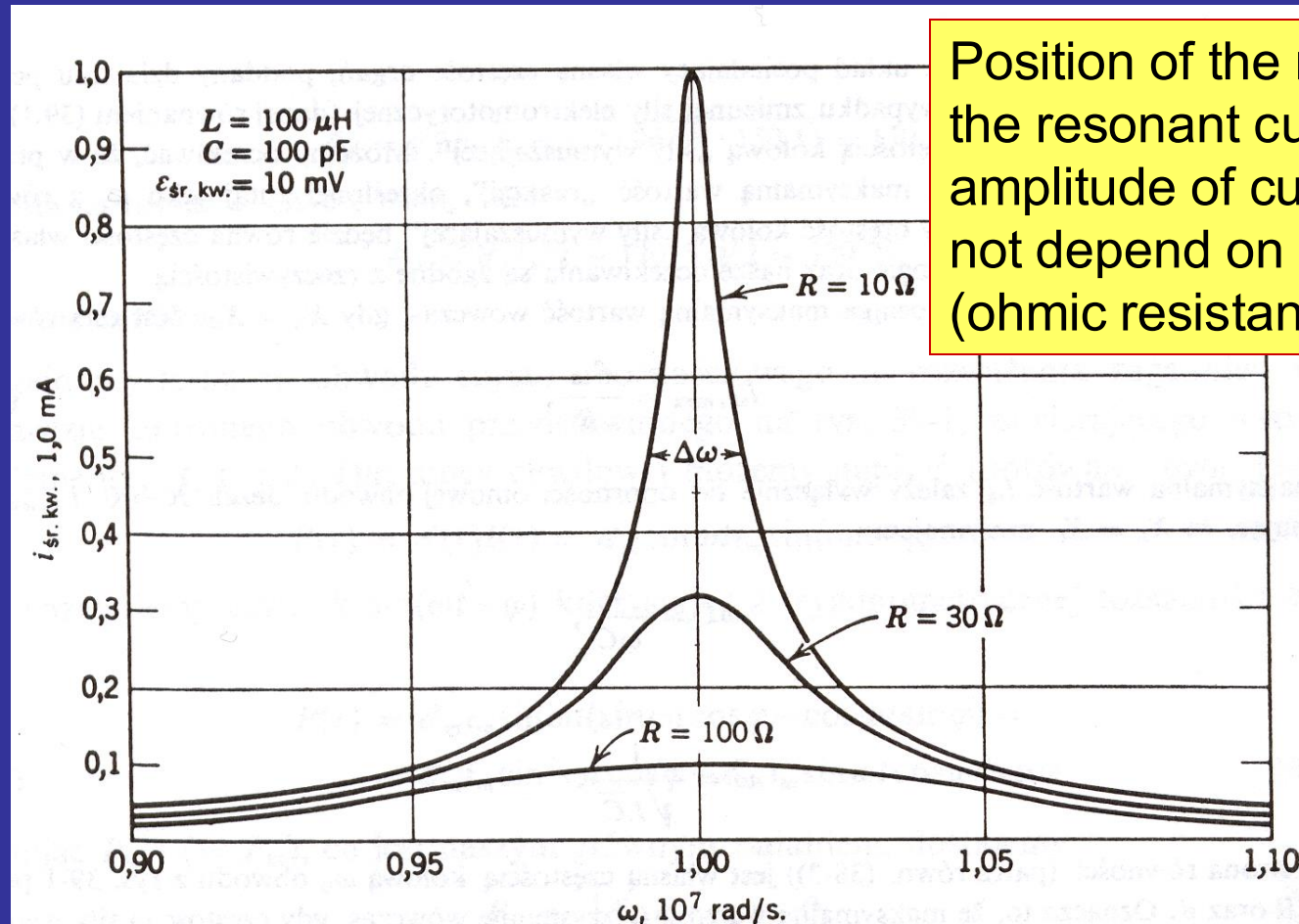
Calculate $\frac{dv_{\max}}{d\omega}$ Show, that the frequency at which there is a maximum of the amplitude of velocity is exactly the natural frequency independently of damping.

When can we observe this type of resonance?

The most frequently in LRC circuits, monitoring the current

Mechanical oscillator	LRC circuit
displacement x	electric charge, q
velocity $v=dx/dt$	current $i=dq/dt$
mass, m	inductance, L
elastic constant, k	Inverse of the capacitance, $1/C$
damping constant, b	resistance, R
force, F	voltage, U

Resonant curve for the amplitude of current



Position of the maximum of the resonant curve of the amplitude of current i_0 does not depend on damping (ohmic resistance) R

Superposition of harmonic oscillations

✓ in the same direction

$$x_1(t) = A_1 \cos(\omega_1 t + \varphi_1)$$

$$x_2(t) = A_2 \cos(\omega_2 t + \varphi_2)$$

$$x_w(t) = x_1(t) + x_2(t)$$

✓ in the perpendicular directions

$$x(t) = A_x \cos(\omega_x t + \varphi_x)$$

$$y(t) = A_y \cos(\omega_y t + \varphi_y)$$

curve $y(x)$

Enhancement and suppression of oscillations in the same direction

Let us assume:

$$x_1(t) = A \cos(\omega t)$$

$$x_2(t) = A \cos(\omega t + \varphi)$$

oscillations of the same amplitude A with the same angular frequency ω , shifted by φ

Result of the superposition:

$$x_{\text{wyp}} = x_1(t) + x_2(t) = 2A \cos \frac{\varphi}{2} \cos(\omega t + \frac{\varphi}{2})$$

Oscillations of the amplitude depending on phase shift φ

$$x_{\text{wyp}} = x_1(t) + x_2(t) = 2A \cos \frac{\varphi}{2} \cos(\omega t + \frac{\varphi}{2})$$

for $\varphi = \pi$, $x_{\text{wyp}} = 0$ complete suppression of oscillations

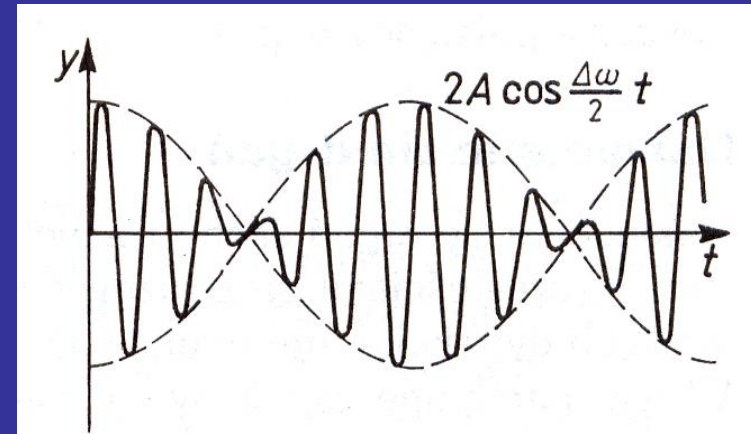
for $\varphi = 2\pi$, $x_{\text{wyp}} = 2A \cos(\omega t)$ amplitude increases by a factor of 2 - enhancement of oscillations

Beats

Superposition of oscillations of the very close frequency

$$y_1(t) = A \sin\left(\omega + \frac{\Delta\omega}{2}\right)t$$

$$y_2(t) = A \sin\left(\omega - \frac{\Delta\omega}{2}\right)t$$



$$y = y_1 + y_2 = A \left[\sin\left(\omega - \frac{\Delta\omega}{2}\right)t + \sin\left(\omega + \frac{\Delta\omega}{2}\right)t \right]$$

From trigonometry:

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

We get:

$$y = 2A \cos\left(\frac{\Delta\omega}{2}t\right) \sin \omega t$$

oscillations of modulated amplitude

Superposition of oscillations in perpendicular directions

Lissajous curves— Jules Antoine Lissajous (1822-1880) first demonstration in 1857

Example 6. Find the result of superposition of the following oscillations:

$$x(t) = A_x \sin \omega t$$

$$y(t) = A_y \sin(\omega t + \varphi)$$

gdy: $\varphi=0$, $\varphi=90^\circ$, $\varphi=180^\circ$

Ellipse is the result
of the superposition
of:

$$x(t) = A_x \sin \omega t$$

$$y(t) = A_y \sin(\omega t + \frac{\pi}{2})$$

Analytic solution:

$$x(t) = A_x \sin \omega t$$

$$y(t) = A_y \sin(\omega t + \frac{\pi}{2}) = A_y \cos(\omega t)$$



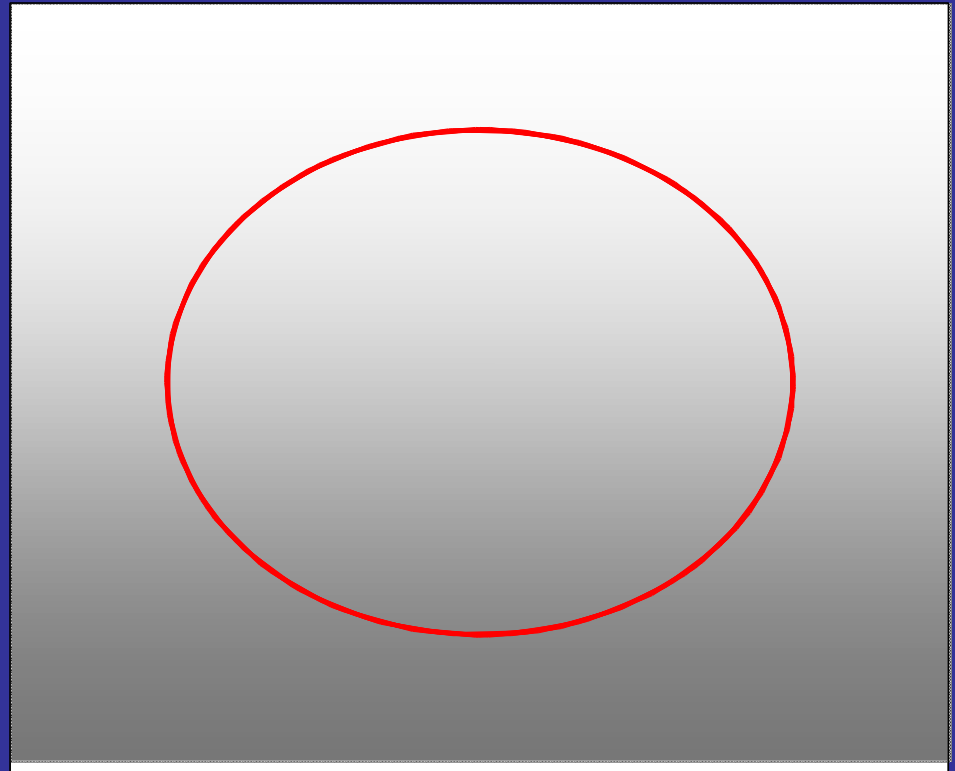
$$\frac{x^2(t)}{A_x^2} = \sin^2 \omega t$$

$$\frac{y^2(t)}{A_y^2} = \cos^2 \omega t$$



$$\frac{x^2(t)}{A_x^2} + \frac{y^2(t)}{A_y^2} = 1$$

Equation of ellipse



SUMMARY

- ❑ Harmonic oscillations: small amplitude and small damping
- ❑ Mechanical energy is conserved without damping
- ❑ Damping leads to the exponential decay of amplitude with time and energy dissipation
- ❑ Driven (forced) oscillator is characterized by the dependence of amplitude on driving frequency that results in the resonance under certain conditions